

SESSION 4

Inflammation Part One

The vascular phase, the cellular phase, and the chemical mediators that run them.

I–II Definitions, general features, and the trilogy of basic changes

III
Blood-vessel reaction in acute inflammation: hemodynamic and permeability changes

IV Exudation: plasma fluid and the leukocyte recruitment cascade

V Chemical mediators: cell-derived and plasma-derived

BUILT FROM

Inflammation (part one) — Prof. Zhou Ren

MBBS Pathology curriculum, ZJU-ISM, Yiwu campus · lecture slides + learning-objectives outline

Robbins Basic Pathology, 11th ed. (chapter 2, Acute and Chronic Inflammation)

*Enriches every mechanism beyond the slides; every such addition is marked **Beyond the slides** in the margin.*

How to use this guide

Part 1 of the inflammation lecture stays entirely inside the blood vessel: how it dilates, leaks, and recruits leukocytes, and which molecules run each step. Morphologic subtypes (serous, fibrinous, suppurative, granulomatous inflammation), systemic manifestations (fever, leukocytosis), and the clinical vocabulary (abscess, sinus, fistula, bacteremia) belong to Part 2 of the lecture, not yet covered here.

Reading the margin rail

A thin rule in the outer margin marks a sidenote: a definition, a unit, or an eponym tied to the paragraph beside it. A **CLINICAL** box is a bedside correlate; an **EXAM TRAP** box is a distinction examiners like to test; a **MNEMONIC** box is a memory aid. Every cascade is drawn as a numbered mechanism chain or a purpose-built diagram, never a photograph standing in for a process it cannot show.

Method

Text extracted from the 95-slide deck and the 22-point learning-objectives outline, restructured around the causal order the lecturer teaches (vascular change first, then leukocyte recruitment, then the molecules behind both), and checked twice against Robbins Basic Pathology 11e for accuracy, completeness, and the current name of every mediator. Two real H&E micrographs are carried over from the lecture slides (credited in `src/img/CREDITS.md`); every mechanism diagram is an original SVG built for this guide.

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01

The Vascular Phase

Inflammation is, before anything else, a story about blood vessels. A tissue with no blood supply cannot inflame at all, only die or adapt; give it a vasculature and it gains a protective response that can dilate, leak, and call in reinforcements. This part follows that story from definition to the leaking vessel wall.

IN THIS PART

- What inflammation is, and why only vascularized tissue can have it
- The three basic changes every inflamed tissue shows, in different proportions
- How the vessel's caliber and wall change in the first minutes to hours
- Why the leaking fluid is an exudate, not a transudate, and why that distinction matters at the bedside

I. Definitions and general features

WORKING DEFINITION

Inflammation is a protective response of vascularized tissue to an injurious agent, one that works through a blood-vessel reaction as its key link and that leads to exudation, the delivery of plasma and leukocytes to the site of injury.

The point of that definition is narrower than it sounds. Inflammation is not simply "the body fighting something off": plants fight off pathogens, and a free-living amoeba can engulf a bacterium, but neither inflames. What both lack is a closed, pressurized circulation. Inflammation is specifically the set of events that a *vascular* system makes possible: a vessel that can widen, that can be made leaky on purpose, and that can hand plasma proteins and white cells to a single point in the tissue.

Inflammation needed a vascular system to exist at all

The lecture's own evolutionary aside is worth keeping, because it is the cleanest way to see why the definition insists on "vascularized tissue." A jellyfish, a plant, or an amoeba facing a hazard has only two options: escape, or engulf. Neither response looks anything like redness, swelling, or pus, because neither has vessels to dilate or a plasma compartment to leak. Once an animal evolves a closed circulation, it gains inflammation as a side effect, and with it a much more powerful, but also more dangerous, defence.

BEYOND THE SLIDES

Robbins frames this as the five cardinal signs Celsus first described (rubor, tumor, calor, dolor, and, added later by Virchow, *functio laesa*), each one a direct readout of the vascular events in this part: redness and heat from hyperemia, swelling from exudate, pain from mediators and pressure on nerve endings.

HIGH YIELD · INFLAMMATION, A DOUBLE-EDGED SWORD

Before antibiotics, a minor bacterial inflammation could kill as reliably as a malignancy; in effect, inflammation *was* neoplasia's rival for how a person died young. The response that protects can also injure: an exaggerated or misdirected inflammatory reaction damages the host's own tissue, which is exactly why a correct diagnosis, not just "there is inflammation," decides whether treatment helps or harms.

II. The trilogy of basic pathologic changes

Every inflamed tissue, whatever the cause, is built from the same three basic changes. They are not three diseases; they are three processes running at once, in a ratio that shifts with time and with the type of inflammation.

TABLE · THE THREE BASIC CHANGES

Change	What happens	Typical cells/tissue involved
Alteration	Degeneration or necrosis of the local inflamed tissue or cell; the direct damage the injurious agent (or the response to it) causes.	Parenchymal cells at the site of injury
Exudation	Intravascular fluid and cells move out into the interstitial space, a body cavity, or a body/mucosal surface.	Plasma and leukocytes crossing the vessel wall
Proliferation	Proliferation or regeneration of the local inflamed tissue or cell, for example macrophages, fibroblasts, lymphocytes, endothelial and epithelial cells.	Stromal and repair-competent cells

CLINICAL CORRELATION · WHY THE RATIO MATTERS

Acute inflammation is dominated by exudation, with alteration and only a little proliferation. Chronic inflammation runs the opposite way: exudation is modest, alteration continues, but proliferation (fibroblasts laying down collagen, new vessels forming) becomes the dominant, tissue-defining feature. Some acute inflammation shows almost no proliferation at all; some shows exudation and nothing else. Reading a biopsy means reading which of the three is winning.

EXAM TRAP

Do not equate "acute" with "exudation only" and "chronic" with "proliferation only": the lecture is explicit that ratios shift, they do not switch off. An acute inflammation can already show early fibroblast proliferation; a chronic inflammation can still show fresh exudation on top of established fibrosis.

Injury → Alteration + Exudation + Proliferation → **Acute pattern:** exudation >> alteration >> proliferation · **Chronic pattern:** proliferation >> exudation

III.1 Hemodynamic changes: vascular caliber and hyperemia



Blood-vessel reaction is always the earliest visible event of inflammation, and it runs in three stages that follow one another within seconds to minutes.

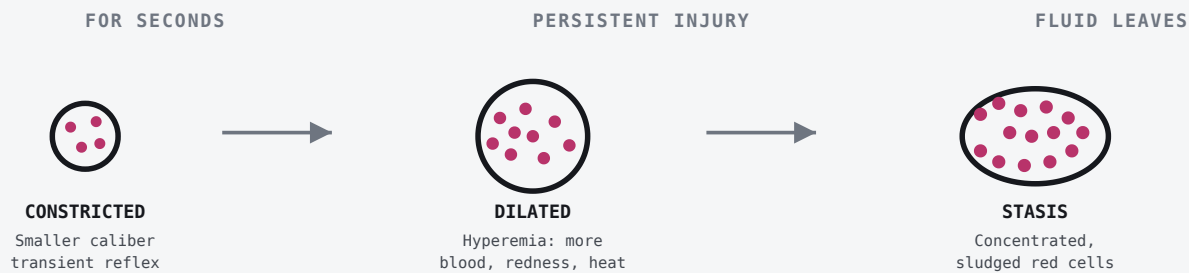


Fig. 1 · Three stages of the caliber change: a brief reflex constriction, then dilation of arterioles and opening of capillary beds (hyperemia: more red cells per vessel, the redness and heat of inflammation), then slowing and sludging of flow (stasis) as fluid leaves through the now-permeable wall and concentrates what remains.

The whole sequence explains four of the five cardinal signs before a single leukocyte has arrived: **redness** and **heat** from the extra, faster-flowing blood of hyperemia, and the start of **swelling** as stasis lets fluid and protein begin to leave the vessel. Pain follows once mediators sensitize nerve endings and once the swelling itself raises local pressure.

TERM
Hyperemia and *congestion* both mean "more blood in a vessel," but hyperemia is the active, arteriolar-driven increase seen here; congestion is the passive, venous-outflow problem seen in heart failure.

III.2 Structural changes in the microvasculature: how the vessel wall leaks

Stasis alone would only slow flow; the leak itself needs a structural change in the endothelial wall. Three distinct mechanisms increase vascular permeability, and they differ in speed, duration, and which vessels they hit, distinctions that examiners like to test directly.

TABLE · THREE MECHANISMS OF INCREASED PERMEABILITY

Mechanism	Trigger	Onset / duration	Vessels affected
Endothelial contraction	Histamine, bradykinin, leukotrienes pulling endothelial cells apart at their junctions	Immediate, transient (15–30 min)	Venules
Direct endothelial injury	Burns, some toxins, severe direct damage killing endothelial cells outright	Immediate and sustained, until the vessel is repaired	Arterioles, capillaries, and venules alike
Increased transcytosis	Growth factors such as VEGF opening channels of fused intracellular vesicles through the endothelial cell	Delayed, can be prolonged	Venules

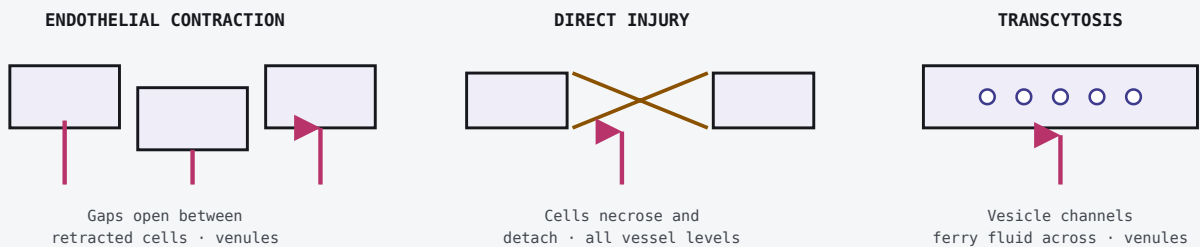


Fig. 2 · The three routes fluid takes across the endothelium. Only endothelial contraction is reversed by antihistamines; the other two are not, which is why severe burns or VEGF-driven leaks do not respond to them.

EXAM TRAP

"Increased hydrostatic pressure" and "increased tissue osmotic pressure" are consequences of the leak, not separate causes of it: once protein-rich exudate raises the interstitial osmotic pressure and stasis raises microcirculatory hydrostatic pressure, more fluid is pulled out, but the leak itself is started by one of the three mechanisms above.

IV.1 Exudation of plasma fluid: Starling forces, transudate vs. exudate

Exudation is defined as the movement of intravascular fluid and cells into the interstitial space, a body cavity, or a body/mucosal surface. Whether that fluid is a transudate or an exudate depends on which side of the Starling equation moved, and the two look and behave differently enough to change management.

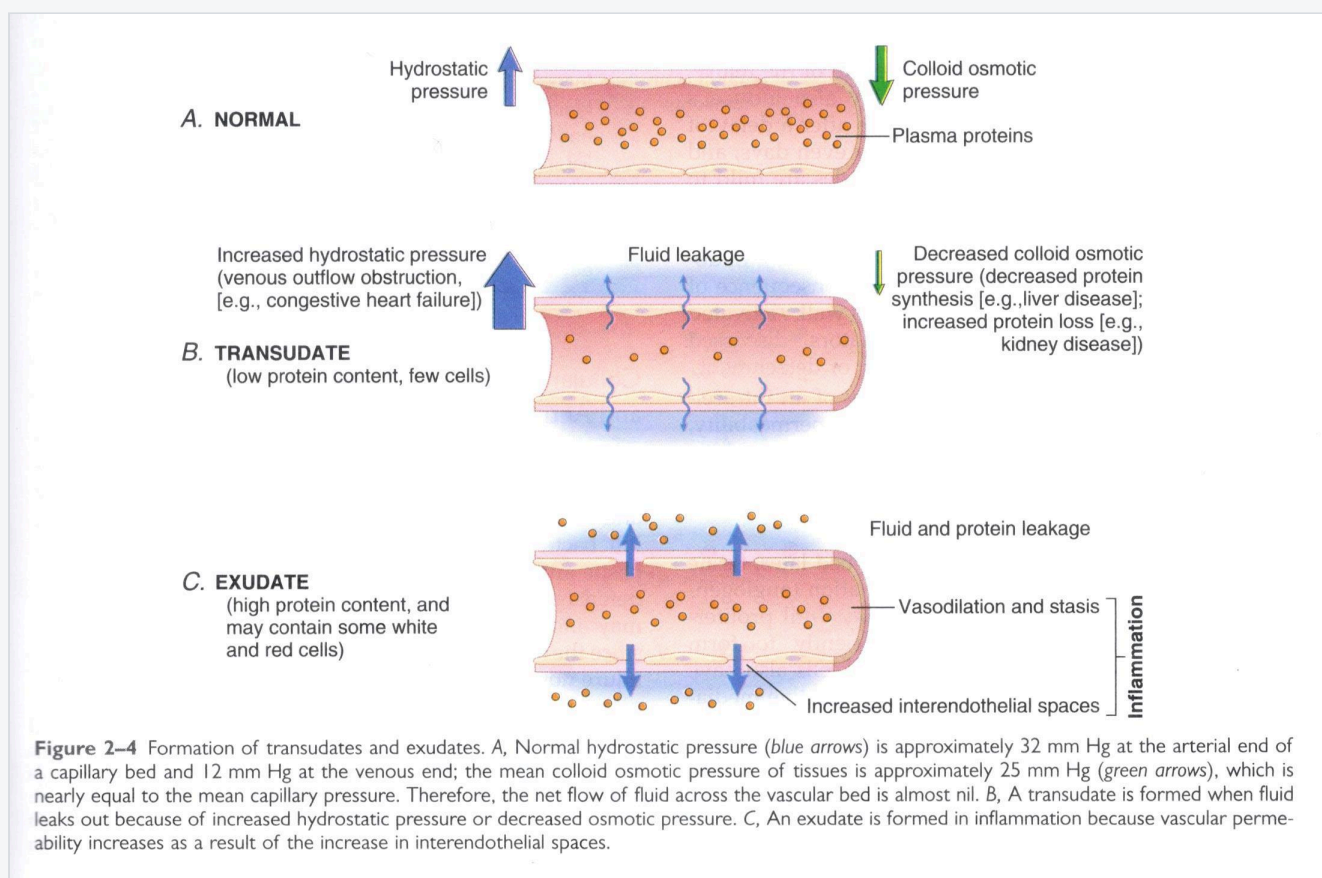


Fig. 3 · Starling forces at the capillary wall. Normally hydrostatic and oncotic pressure roughly balance. A transudate forms when that balance is disturbed by pressure or protein changes elsewhere in the body, with the vessel wall itself intact. An exudate forms in inflammation because the wall itself becomes leaky, letting protein and cells out alongside the fluid.

TABLE · TRANSUDATE VS. EXUDATE, THE DIFFERENTIAL FEATURES

Feature	Transudate	Exudate
Mechanism	Hemodynamic imbalance; vessel wall intact	Increased vascular permeability; inflammation
Protein content	Low	High
Specific gravity	< 1.012	> 1.020
Cell content	Few	May be numerous (leukocytes, sometimes red cells)
Typical causes	Congestive heart failure, liver disease, kidney disease/nephrotic syndrome	Infection, any acute inflammatory process

MNEMONIC

"Ex" leaks EXtra: an EXudate has EXtra protein and EXtra cells, because the vessel wall is EXplicitly damaged. A transudate simply TRANSits through an intact wall pushed by pressure.

DEFINITION

Edema: an excess of fluid in the interstitial tissue or serous cavities; it can be either a transudate or an exudate, so "edema" alone does not tell you the mechanism.

02

The Cellular Phase and Its Molecules

A leaking vessel only delivers plasma. Getting a neutrophil out of the blood and onto a bacterium takes a four-step cascade of adhesion molecules, a chemical compass, and then a killing machine, all run by a short list of named receptors and mediators that examiners test individually.

IN THIS PART

- Margination, rolling, adhesion, and transmigration, and the named molecule at each step
- Chemotaxis: how a leukocyte reads a chemical gradient and moves along it
- Phagocytosis and killing, oxygen-dependent and oxygen-independent
- Which leukocyte dominates at which stage, and the macrophage's disguises
- Cell-derived and plasma-derived chemical mediators

IV.2 Inflammatory cell infiltration: overview

DEFINITION

Inflammatory infiltration is the process by which leukocytes traverse the vessel wall and enter the interstitial space to carry out phagocytosis during inflammation. Once emigrated, these cells are called **inflammatory cells**.

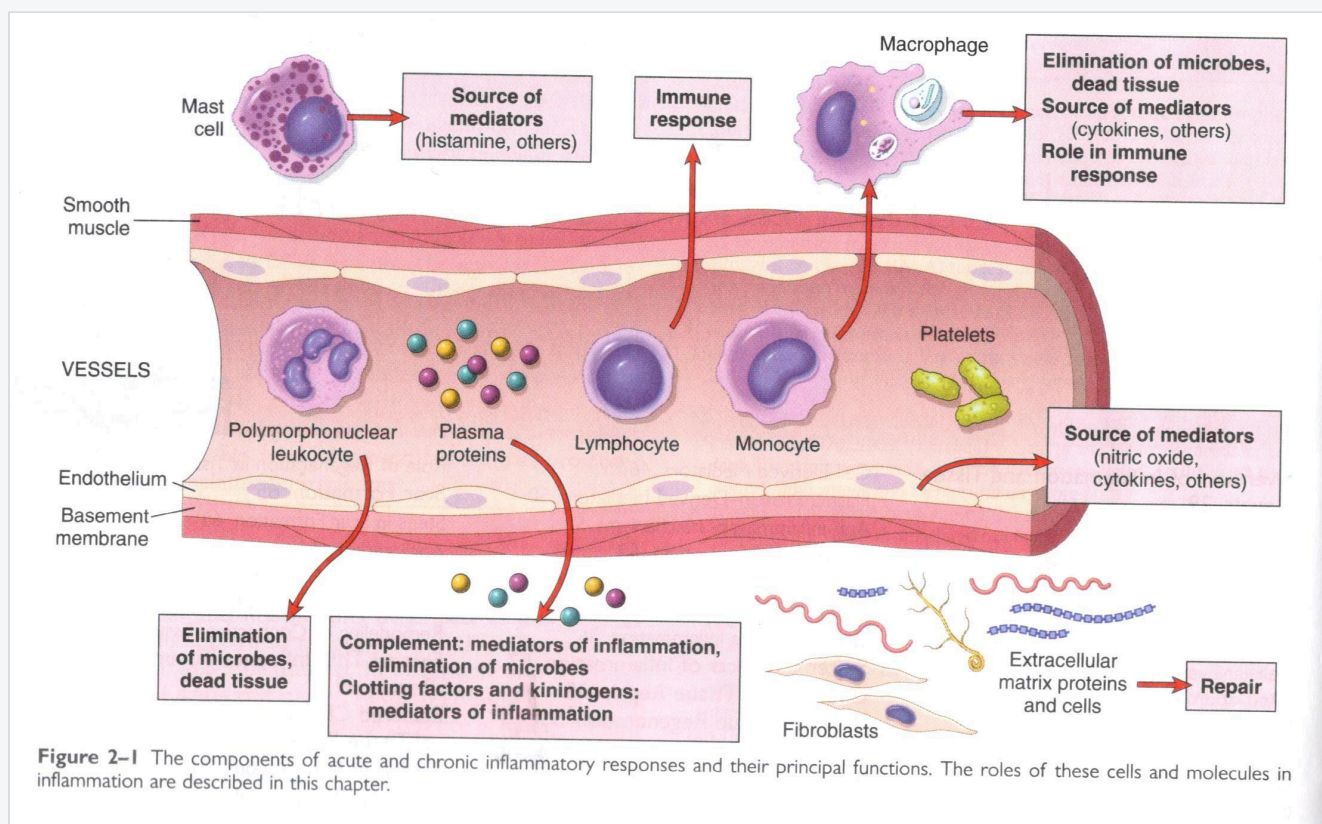


Fig. 4 · The cast of this part. Everything this cellular phase adds sits inside the same vessel wall shown in Part One: leukocytes and plasma proteins that cross it, and the cells and matrix that receive them on the other side.

Four named steps carry a leukocyte from the axial stream of flowing blood to the site of injury, followed by chemotaxis to steer it and phagocytosis to let it act. Each step is executed by its own set of adhesion molecules, and each set is upregulated only when the step before it has happened, which is why the sequence is obligatory and cannot skip ahead.

(a) Margination and pavement

As stasis slows flow, red cells aggregate into larger masses (rouleaux) that push leukocytes out of the fast-moving central (axial) stream and toward the vessel wall. Leukocytes then tumble along the endothelial surface

and line up against it, a process the lecture calls **pavementing**: the wall looks paved with white cells under the microscope.

(b) Rolling and firm adhesion

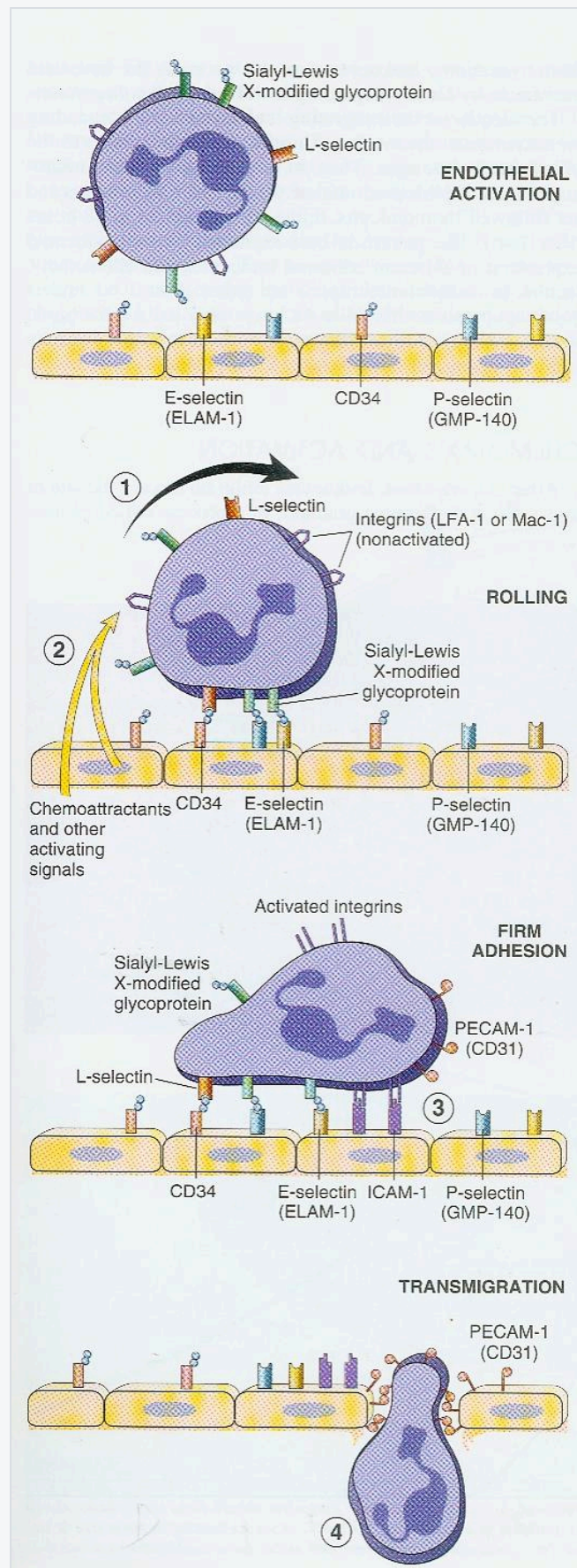


Fig. 5 · The adhesion cascade. Selectins make the leukocyte roll instead of flow past; endothelium-displayed chemokines then activate the leukocyte's integrins, which bind Ig-superfamily ligands and arrest it; PECAM-1 finally guides it, homophilically, through the endothelial junction and its basement membrane.

TABLE · MOLECULE MASTER LIST

Family	On the leukocyte / endothelium
Selectins	L-selectin (leukocyte); E-selectin/ELAM-1 and P-selectin/GMP-140 (endothelium), binding sialyl-Lewis X
Integrins	LFA-1, Mac-1 on the leukocyte, activated by chemokines, binding ICAM-1/VCAM-1 on endothelium
Ig-superfamily	ICAM-1, VCAM-1 (integrin ligands); PECAM-1/CD31 drives transmigration itself
Chemokines	Displayed on the endothelial surface, convert rolling into firm adhesion

○ WHY THE SEQUENCE CANNOT SKIP A STEP

- Selectins bind weakly and reversibly to their carbohydrate ligand, so the leukocyte only slows and rolls; it cannot yet stop.

WHY: rolling samples the endothelium long enough for chemokine signalling to occur.

- Chemokines held on the endothelial surface trigger a G-protein-coupled signal inside the rolling leukocyte that switches its integrins from low- to high-affinity.

WHY: integrins are present at baseline but non-adhesive until activated; this is the rate-limiting checkpoint of the whole cascade.

- High-affinity integrins bind ICAM-1/VCAM-1 tightly: the leukocyte arrests completely (firm adhesion).

- PECAM-1 on the leukocyte binds PECAM-1 on the endothelial junction (homophilic binding) and pulls the cell through; collagenases then let it cross the basement membrane.

NET RESULT a leukocyte physically inside the tissue, still needing a compass to find the injury.

EXAM TRAP: NAME THE DEFICIENCY

Leukocyte adhesion deficiency type 1 (LAD-1) is a defect in the integrin $\beta 2$ subunit (CD18, shared by LFA-1 and Mac-1): leukocytes roll normally but never arrest, so patients get recurrent bacterial infections with delayed wound healing and, oddly, very little visible pus.

(c) Chemotaxis and transmigration

DEFINITION

Chemotaxis is the emigration of a leukocyte toward the site of a chemical stimulus; a **chemotactic agent** is any chemical substance that can induce that emigration.

TABLE · CHEMOTACTIC AGENTS

Class	Examples
Exogenous	Soluble bacterial products, classically N-formyl-methionine peptides (fMLP)
Endogenous	Complement component C5a; leukotriene B4 (LTB4); IL-8 and other chemokines

All of these bind specific seven-transmembrane, G-protein-coupled receptors on the leukocyte surface, triggering an intracellular cascade of phospholipid metabolism that culminates in a rise in intracellular calcium. That calcium rise assembles the cytoskeletal contractile elements (actin, myosin) the cell needs to extend a pseudopod and crawl up the gradient.

○ CHEMOTACTIC SIGNAL TRANSDUCTION, IN ORDER

- 2 Chemoattractant binds its receptor; an inactive G-protein is switched on.
- 3 Active phospholipase C splits membrane PIP₂ into diacylglycerol (DAG) and inositol trisphosphate (IP₃).
- 4 IP₃ releases calcium from the endoplasmic reticulum; DAG activates protein kinase C.
- 5 Rising calcium and protein kinase C together assemble actin/myosin filaments (motility, pseudopod formation), increase the number and affinity of adhesion molecules, and trigger the oxidative burst and granule secretion.

WHY: *the same calcium signal that lets the cell move is what arms it to kill once it arrives.*

NET RESULT one signal (a calcium rise) both steers the cell and arms its killing machinery.

(d) Leukocyte actions: phagocytosis, degradation, immunity, tissue injury

Once a leukocyte has arrived, it has three possible jobs: eat and destroy the offending agent (phagocytosis and degradation), present it to trigger adaptive immunity, or, as an unwanted side effect of doing either, injure the host's own tissue.

DEFINITION

Phagocytosis is the process by which a leukocyte engulfs a pathogenic microorganism, a foreign body, or tissue debris at the inflammatory focus.

TABLE · THE THREE PHAGOCYTES

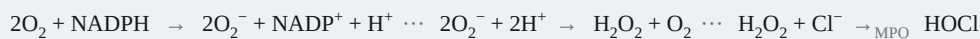
Phagocyte	Also called	What it clears
Neutrophil	Microphage	Many kinds of pathogenic microorganism and tissue debris; the early responder
Macrophage	Large phagocyte	Larger debris, more organisms per cell, sustained clearance and antigen presentation
Eosinophil	—	Parasites (via granule proteins), and a role in allergic reactions

○ PHAGOCYTOSIS AND DEGRADATION, IN ORDER

- 2 **Recognition and attachment.** The phagocyte's receptors recognize the particle directly, or recognize it after it has been coated by an **opsonin**, a serum protein, chiefly IgG or C3b, that makes the target far easier to bind.
WHY: opsonization converts a slippery target into one the phagocyte's receptors grip efficiently, which is why opsonin (or complement) deficiency causes recurrent infection even with a normal phagocyte count.
- 3 **Engulfment.** The phagocyte's membrane extends pseudopods around the particle, and they fuse to close it inside a membrane-bound vacuole, the **phagosome**.
- 4 The phagosome fuses with a lysosome to form a **phagolysosome**, with degranulation of lysosomal contents into it.
- 5 **Killing or degradation**, by an oxygen-dependent and an oxygen-independent mechanism running side by side.

TABLE · KILLING MECHANISMS

Mechanism	How it works
Oxygen-dependent	Phagocytosis triggers an oxidative (respiratory) burst : a sudden rise in oxygen consumption and glycogen catabolism generates reactive oxygen species (ROS). NADPH oxidase converts O_2 to superoxide (O_2^-), which is then converted to hydrogen peroxide (H_2O_2); myeloperoxidase (MPO) then uses H_2O_2 plus a halide to generate hypochlorous acid (HOCl), a powerful microbicidal oxidant.
Oxygen-independent	Granule enzymes that do not need oxygen: lysozyme, lactoferrin, defensins, and other bactericidal proteins.



CLINICAL CORRELATION

Chronic granulomatous disease is a defect of NADPH oxidase: phagocytes engulf bacteria normally but cannot generate the oxidative burst, so patients suffer recurrent infection with catalase-positive organisms (*Staphylococcus aureus*, *Aspergillus*) that would otherwise destroy their own H_2O_2 and be killed by it.

A LITTLE DICTIONARY

NADPH: coenzyme II. O_2^- : superoxide anion. H_2O_2 : hydrogen peroxide. MPO: myeloperoxidase. HOCl: hypochlorous acid. ROS: reactive oxygen species.

HIGH YIELD · THE OTHER TWO JOBS

Immunity: macrophages and dendritic cells process and present antigen to lymphocytes, linking innate inflammation to the adaptive immune response. **Tissue injury:** the same enzymes and ROS that kill microbes are released into the extracellular space and damage normal host tissue, which is the mechanistic basis of collateral damage in prolonged or misdirected inflammation.

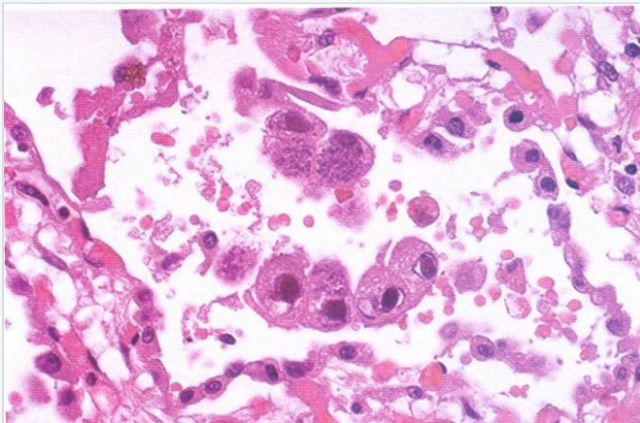
· Cell-type significance and the macrophage's morphologic disguises

Which leukocyte dominates a lesion is itself diagnostic information, not just descriptive detail.

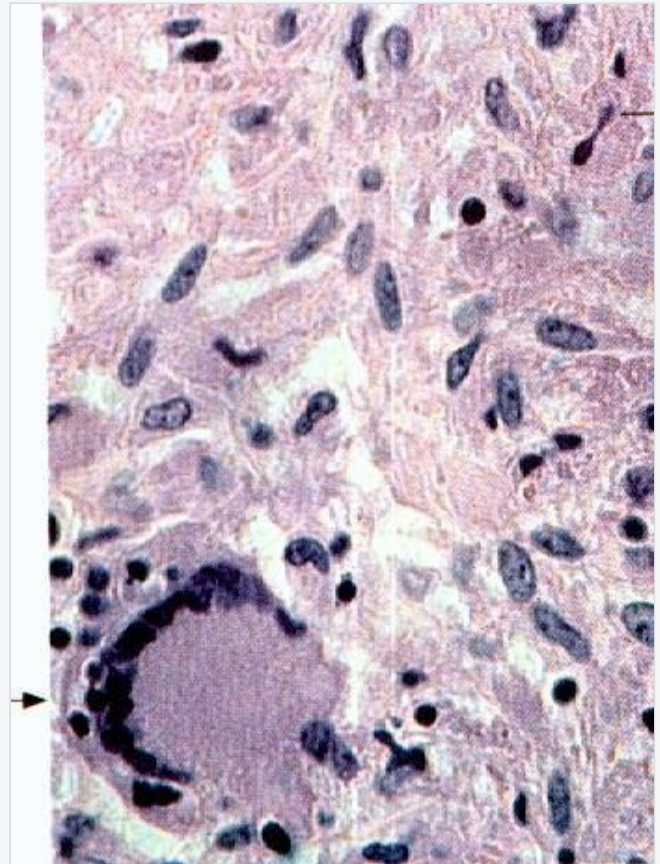
TABLE · PATHOLOGICAL AND CLINICAL SIGNIFICANCE OF EACH INFLAMMATORY CELL

Cell	Read as
Neutrophil	Early stage of acute inflammation; purulent (suppurative) inflammation
Monocyte	Later stage of acute inflammation; some specific infectious diseases
Eosinophil	Mainly allergy and parasite infection
Lymphocyte	Viral infection and chronic inflammation; also characteristic of syphilis

The macrophage in particular changes shape with its job. Four morphologic forms recur across pathology and are worth recognizing on sight.



MICRO **Fig. 6** H&E, inflammatory focus with epithelioid macrophages and an early multinucleated giant cell forming by fusion.



MICRO **Fig. 7** Langhans' giant cell (nuclei arranged in a horseshoe at the cell's rim) beside sheets of epithelioid cells, H&E.

TABLE · THE MACROPHAGE'S FOUR DISGUISES

Form	Looks like	Seen in
Epithelioid cell	Elongated, pink, epithelium-like cytoplasm; poor at phagocytosis, good at secretion	Granulomas
Langhans' giant cell	Many nuclei arranged in a horseshoe or ring at the cell periphery	Classically tuberculous granulomas
Foreign-body giant cell	Nuclei scattered randomly throughout the cytoplasm	Around inert foreign material (suture, splinter)
Foamy cell	Abundant, vacuolated, lipid-laden cytoplasm	Atherosclerosis, xanthomas, sites of lipid breakdown

EXAM TRAP

Nucleus arrangement is the whole distinction between the two giant cells: **horseshoe at the rim = Langhans**; **scattered randomly = foreign-body**. Both are fused epithelioid macrophages; neither is a distinct cell lineage.

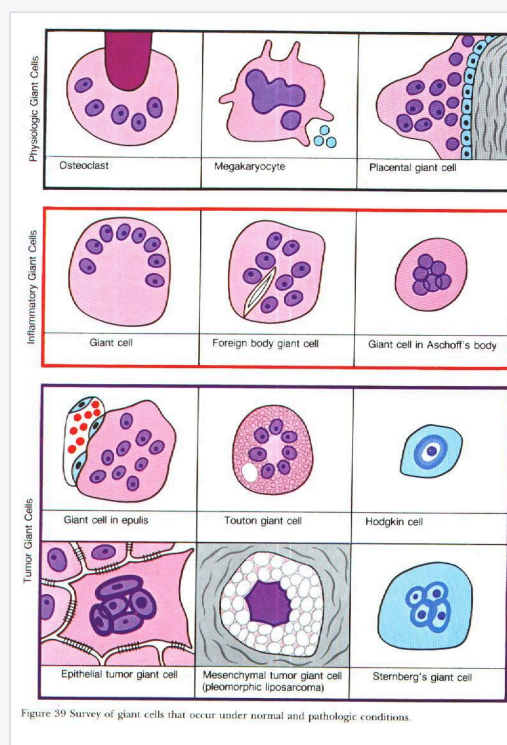


Fig. 8 · The giant-cell family in full: physiologic (osteoclast, megakaryocyte), inflammatory (foreign-body, Langhans'), and tumor giant cells, side by side. Same fusion mechanism, different trigger.

V.1 Chemical mediators: cell-derived

DEFINITION

An **inflammatory mediator** is a chemical substance that mediates the occurrence and progression of inflammation. Mediators are **exogenous** (bacteria and their products) or **endogenous**, and endogenous ones are further split by origin into **cell-derived** and **plasma-derived**.

BEYOND THE SLIDES

Robbins notes granuloma formation itself (the organized collection of epithelioid macrophages) is a Part 2 topic; this section only covers the macrophage's own shape-shifting, which the lecture introduces here as part of leukocyte biology.

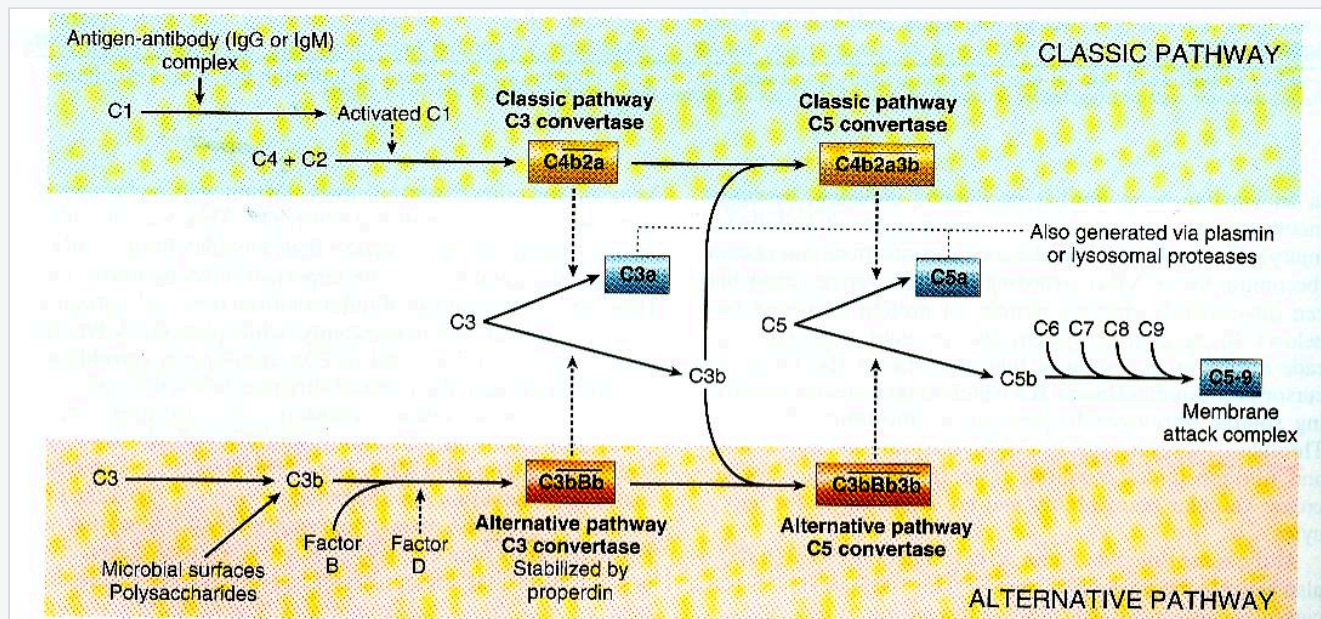


Figure 2-11

Overview of complement activation pathways. The classic pathway is initiated by C1 binding to antigen-antibody complexes; the alternative pathway is initiated by C3b binding to various activating surfaces, such as microbial cell walls. The C3b involved in alternative pathway initiation may be generated in several ways, including spontaneously, by the classic pathway, or by the alternative pathway itself (see text). Both pathways converge and lead to the formation of inflammatory complement mediators (C3a and C5a) and the membrane attack complex. Bars over the letter designations of complement components indicate enzymatically active forms; dashed lines indicate proteolytic activities of various components. (Modified from Abbas AK, et al: Cellular and Molecular Immunology, 2nd ed. Philadelphia, WB Saunders, 1994.)

Fig. 9 · Where mediators come from. Cell-derived mediators sit preformed in granules or are synthesized on demand; plasma-derived mediators arrive as inactive precursors and need Factor XII or complement activation to switch on.

Vasoactive amines: the first responders

Histamine is stored preformed in mast-cell and basophil granules (and in platelets) and is released within seconds of a stimulus, causing arteriolar dilation and increased venular permeability by endothelial contraction; it is inactivated quickly by histaminase, which is why its effect is so short-lived. **Serotonin (5-hydroxytryptamine, 5-HT)** is stored in platelets and in argyrophil cells of the gastrointestinal tract and produces a similar effect.

Arachidonic acid metabolites: prostaglandins and leukotrienes

Arachidonic acid (AA) is cleaved from membrane phospholipids by phospholipase A2 and metabolized down one of two enzymatic routes, each blocked by a different, clinically important class of drug.

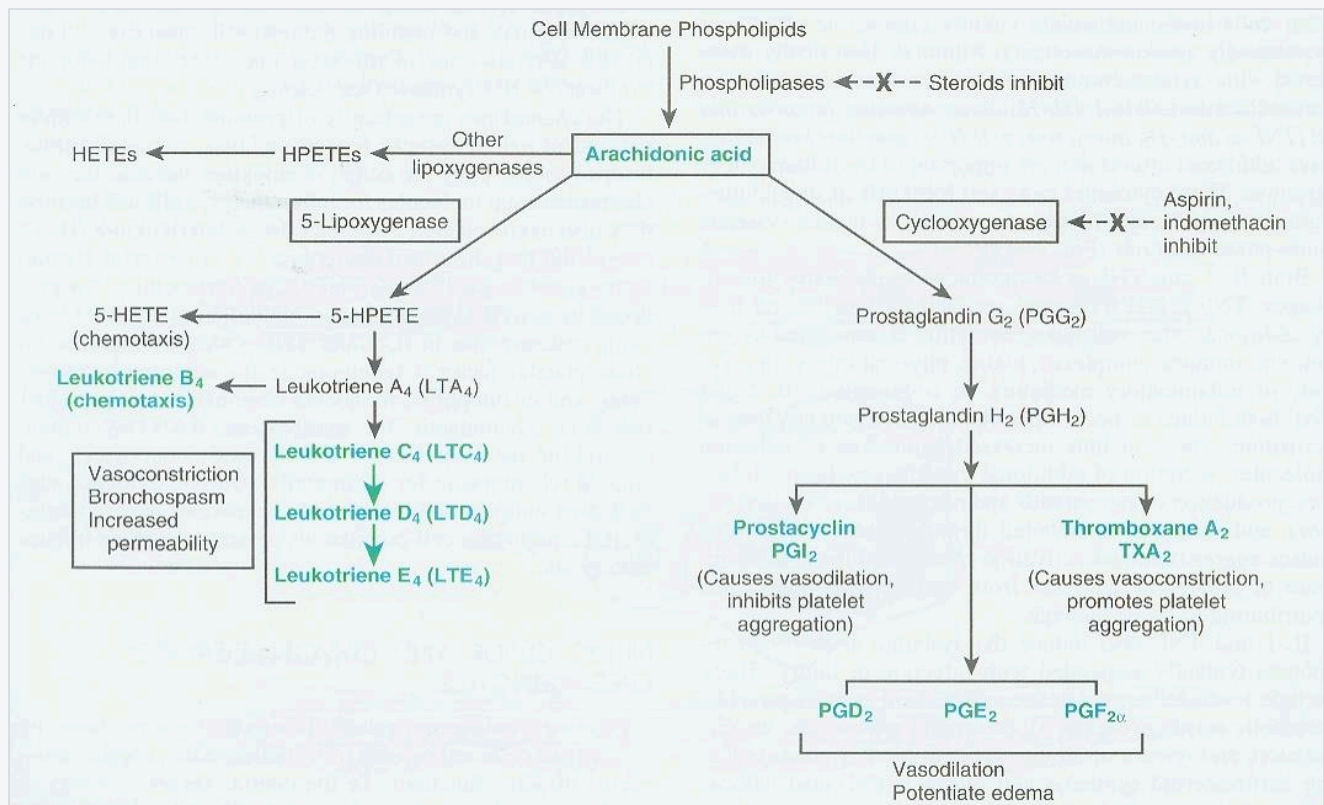


Figure 2–12

Generation of arachidonic acid metabolites and their roles in inflammation.

Fig. 10 · Arachidonic acid metabolism. Steroids act upstream, blocking phospholipase A₂ and therefore both branches at once; aspirin and other NSAIDs block only cyclooxygenase, leaving the lipoxygenase branch (and leukotriene production) untouched, which is why NSAID-sensitive asthma can worsen with aspirin (unblocked arachidonic acid is shunted toward leukotrienes).

COX has two isoforms: COX-1 (constitutive, gastric/renal protection) and COX-2 (inducible at sites of inflammation), which is why selective COX-2 inhibitors spare the stomach.

EXAM TRAP

Do not say "aspirin blocks arachidonic acid production": it blocks only cyclooxygenase, one branch of two. Steroids are the ones that block phospholipase A₂ and shut down both branches.

Cytokines, PAF, and nitric oxide

TABLE · REMAINING CELL-DERIVED MEDIATORS

Mediator	Source	Action
Cytokines (TNF-α, IL-1, IL-6)	Polypeptides, chiefly from activated lymphocytes and macrophages	Modulate the function of other cells; systemic effects (fever, acute-phase response)
Platelet-activating factor (PAF)	A phospholipid-derived mediator from leukocytes, endothelium, and platelets	Aggregates platelets and causes their degranulation
Nitric oxide (NO)	Endothelium and macrophages	Vasodilation; microbicidal in activated macrophages

V.2 Chemical mediators: plasma-derived

Three plasma protein systems, complement, kinin, and coagulation, are not independent: all three are triggered together when activated Factor XII (Hageman factor) meets collagen, basement membrane, or activated platelets at a site of injury.

BEYOND THE SLIDES

The lecture's own aside on IL-6 is worth keeping as written: interleukin-6 drew intense attention during COVID-19 because a high IL-6 level correlated with poor prognosis, which is the rationale behind IL-6-receptor-blocking drugs (tocilizumab) used in severe cases. A 2020 meta-analysis of 11 studies confirmed the correlation the lecture describes: serum IL-6 was significantly higher in severe/critical COVID-19 than in mild/moderate disease (mean difference +23.1 pg/mL, independent of age and sex; Mojtabavi et al., *Eur Cytokine Netw* 2020, PMID 32933891).

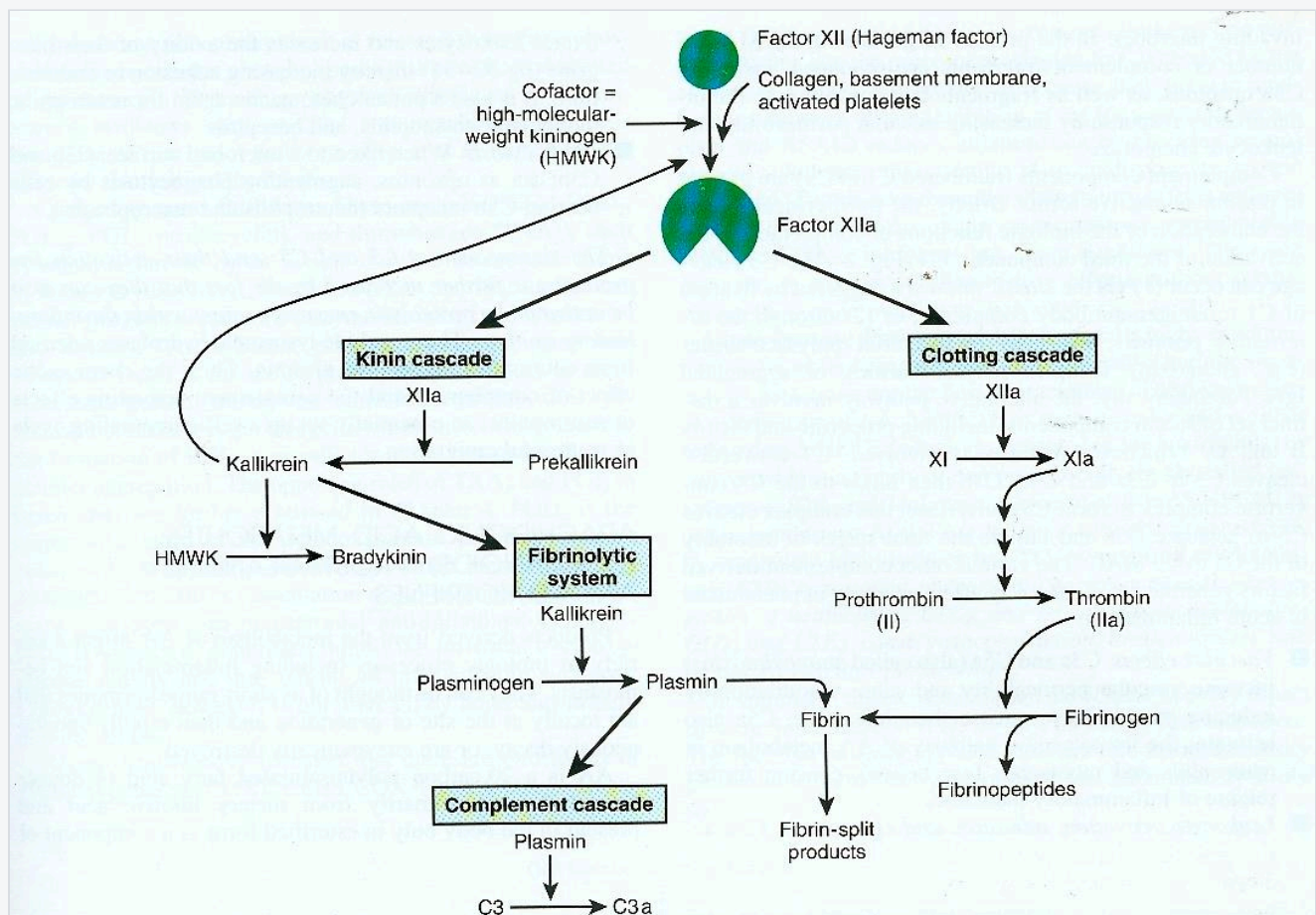


Figure 2-10

Plasma mediator systems triggered by activation of factor XII (Hageman factor): their interrelationships and major effects.

Fig. 11 · One trigger, three cascades. Factor XIIa starts the kinin, complement, and clotting systems together, and they cross-activate each other (kallikrein can activate C3 directly; plasmin from the fibrinolytic system can too), which is why a single injury produces a mixed mediator picture rather than one clean pathway.

TABLE · COMPLEMENT'S TWO ACTIVATION ROUTES

Pathway	Trigger	Shared endpoint
Classical	Antigen–antibody (IgG or IgM) complex activates C1	C3 convertase → C3a + C3b; C5 convertase → C5a + C5b; C5b initiates the membrane attack complex (C5b-9), which lyses the target cell
Alternative	C3b binds microbial surfaces/polysaccharides directly, stabilized by properdin	

HIGH YIELD · C3A AND C5A

C3a and C5a are **anaphylatoxins**: they trigger mast-cell histamine release and so increase vascular permeability indirectly; C5a is additionally a direct chemotactic agent for neutrophils, and C3b is an opsonin. One family of proteins therefore contributes to permeability, chemotaxis, and phagocytosis all at once.

TABLE · MASTER MEDIATOR TABLE (MOST LIKELY MEDIATOR PER EFFECT)

Effect	Mediators
Vasodilation	Prostaglandins, nitric oxide
↑ vascular permeability	Vasoactive amines; C3a and C5a (via amine release); bradykinin; leukotrienes C4, D4, E4; PAF
Chemotaxis, leukocyte activation	C5a; leukotriene B4; bacterial products; chemokines (e.g. IL-8)
Fever	IL-1, IL-6, TNF-α; prostaglandins
Pain	Prostaglandins; bradykinin
Tissue damage	Neutrophil and macrophage lysosomal enzymes; oxygen metabolites; nitric oxide

REVIEW

R Self-Test, Tables, Glossary

Fifteen questions in the single-best-answer format used on the MBBS pathology exam, an answer key with the reasoning for each distractor, a consolidated set of master tables for last-minute review, and a bilingual glossary.

Self-test question bank

Q1

DEFINITIONS

A tissue with no blood supply is injured and reacts by escape or engulfment only, never by redness or swelling. This best illustrates that inflammation requires:

- A. A functioning adaptive immune system
- B. A vascularized tissue
- C. Intact innervation
- D. A population of resident macrophages

Q2

BASIC CHANGES

A biopsy shows extensive fibroblast proliferation and new vessel formation with minimal fresh exudate. This pattern is most typical of:

- A. Early acute inflammation
- B. Chronic inflammation
- C. Transudation
- D. Anaphylaxis

Q3

HEMODYNAMICS

Which best describes the correct order of the vascular caliber change in acute inflammation?

- A. Dilation, then transient constriction, then stasis
- B. Transient constriction, then dilation, then stasis
- C. Stasis, then dilation
- D. Constriction throughout

Q4

PERMEABILITY

A burn victim's severely damaged skin leaks plasma from arterioles, capillaries, and venules alike, immediately and for days afterward. The mechanism is:

- A. Endothelial contraction
- B. Increased transcytosis
- C. Direct endothelial injury
- D. Leukocyte-mediated injury only

Q5

TRANSUDATE/EXUDATE

Pleural fluid with specific gravity 1.030 and numerous neutrophils is best classified as:

- A. Transudate
- B. Exudate
- C. Lymph
- D. Chyle

Q6**ADHESION CASCADE**

A child has recurrent bacterial infections, delayed umbilical cord separation, and neutrophilia with almost no pus at infection sites. Flow cytometry shows absent CD18. The step of leukocyte recruitment that fails is:

- A. Rolling (selectins)
- B. Firm adhesion (integrins)
- C. Transmigration (PECAM-1)
- D. Chemotaxis (receptors)

Q7**CHEMOTAXIS**

The second-messenger rise that both arms a neutrophil's cytoskeleton for movement and triggers its oxidative burst is:

- A. cAMP
- B. Intracellular calcium
- C. Sodium influx
- D. Extracellular potassium

Q8**PHAGOCYTOSIS**

A boy has recurrent infections with catalase-positive organisms; his neutrophils phagocytose bacteria normally but the nitroblue-tetrazolium test is abnormal. The missing enzyme system is:

- A. Lysozyme
- B. NADPH oxidase
- C. Myeloperoxidase only
- D. Lactoferrin

Q9**GIANT CELLS**

A granuloma shows a multinucleated cell with nuclei arranged in a peripheral horseshoe. This is best termed a:

- A. Foreign-body giant cell
- B. Langhans' giant cell
- C. Touton giant cell
- D. Osteoclast

Q10**AA PATHWAY**

A patient with aspirin-exacerbated respiratory disease develops bronchospasm after taking ibuprofen. The mechanistic explanation is that COX inhibition:

- A. Directly triggers histamine release from mast cells
- B. Shunts arachidonic acid toward the lipoxygenase pathway, increasing leukotrienes
- C. Increases thromboxane A₂ directly
- D. Activates complement

Q11 PLASMA CASCADES

Factor XIIa (activated Hageman factor) directly initiates all of the following EXCEPT:

- A. The kinin cascade
- B. The clotting cascade
- C. Antibody class switching
- D. The fibrinolytic system, which can cross-activate complement

Q12 COMPLEMENT

A complement fragment that is both an anaphylatoxin and a direct neutrophil chemoattractant is:

- A. C3b
- B. C5a
- C. C1q
- D. C5b-9

Q13 MEDIATORS, FEVER

Which pair of mediators is most responsible for the fever of acute inflammation?

- A. Histamine and serotonin
- B. IL-1 and prostaglandins
- C. Bradykinin and C5a
- D. PAF and nitric oxide

Q14 OPSONIZATION

The two most important opsonins are:

- A. IgM and C1q
- B. IgG and C3b
- C. Albumin and fibrinogen
- D. IL-8 and LTB₄

Q15 INTEGRATION

Put the following in the correct causal order: (1) chemotaxis, (2) firm adhesion, (3) margination, (4) rolling, (5) transmigration.

- A. 3, 4, 2, 5, 1
- B. 4, 3, 2, 1, 5
- C. 3, 2, 4, 5, 1
- D. 1, 3, 4, 2, 5

Answer key

Q1 **B** A vascularized tissue.

Inflammation is defined by a blood-vessel reaction; without vessels there is no hyperemia, no exudation, only the primitive alternatives of escape or engulfment.

Q2 B Chronic inflammation.

Proliferation (fibroblasts, new vessels) dominating over exudation is the chronic pattern; acute inflammation is exudation-dominant.

Q3 B Transient constriction, then dilation, then stasis.

The lecture's own order: seconds of reflex constriction, then hyperemia as arterioles and capillary beds open, then slowing/sludging as fluid leaves.

Q4 C Direct endothelial injury.

Only direct injury is immediate AND sustained AND hits every vessel level; endothelial contraction is venule-only and transient, transcytosis is delayed and venule-only.

Q5 B Exudate.

Specific gravity above 1.020 plus numerous neutrophils both point to a protein-rich, cell-rich exudate from an inflamed pleura, not a hemodynamic transudate.

Q6 B Firm adhesion (integrins).

CD18 is the shared $\beta 2$ subunit of LFA-1 and Mac-1; its absence is leukocyte adhesion deficiency type 1. Rolling (selectins) is intact, so neutrophils reach the wall but never arrest, explaining neutrophilia with little visible pus.

Q7 B Intracellular calcium.

The chemoattractant-receptor-G-protein-PLC-IP₃ cascade culminates in a calcium rise that assembles actin/myosin for movement and triggers degranulation/oxidative burst simultaneously.

Q8 B NADPH oxidase.

This is chronic granulomatous disease: phagocytosis is normal (recognition/engulfment intact) but the oxidative burst fails, so catalase-positive organisms that destroy their own H₂O₂ survive.

Q9 B Langhans' giant cell.

Peripheral horseshoe/ring arrangement of nuclei defines the Langhans' giant cell, classically in tuberculous granulomas; foreign-body giant cells scatter nuclei randomly.

Q10 B Shunts arachidonic acid toward the lipoxygenase pathway, increasing leukotrienes.

Blocking cyclooxygenase leaves more substrate available for 5-lipoxygenase, raising bronchoconstrictive, permeability-increasing leukotrienes (C₄/D₄/E₄) and LTB₄.

Q11 C Antibody class switching.

Class switching is a germinal-center, cytokine-driven adaptive immune event, unrelated to Factor XIIa; the other three are exactly the plasma cascades XIIa triggers together.

Q12 B C5a.

C3a and C5a are both anaphylatoxins, but only C5a is also a direct neutrophil chemoattractant; C3b is an opsonin, not a chemoattractant.

Q13 B IL-1 and prostaglandins.

IL-1, IL-6, and TNF- α act on the hypothalamus to raise its set point, and prostaglandin E₂ is the direct mediator of that reset, which is why antipyretics (COX inhibitors) lower fever.

Q14 B IgG and C3b.

These are the two opsonins named explicitly in the phagocytosis mechanism; they coat the target so Fc and complement receptors on the phagocyte can bind it efficiently.

Q15 A 3, 4, 2, 5, 1 (margination, rolling, firm adhesion, transmigration, chemotaxis).

Chemotaxis is the compass that only matters once the leukocyte is already through the wall; every adhesion step before it is a fixed, obligatory order.

Master tables, quick reference

Condensed recap of every table already built into the text, gathered here for a last pass before the exam.

QUICK TABLE · THE ADHESION-MOLECULE PAIRS

Step	Leukocyte molecule	Endothelial ligand
Rolling	L-selectin, sialyl-Lewis X	E-selectin (ELAM-1), P-selectin (GMP-140)
Firm adhesion	LFA-1, Mac-1 (integrins)	ICAM-1, VCAM-1
Transmigration	PECAM-1 (CD31)	PECAM-1 (CD31), homophilic

QUICK TABLE · TRANSUDATE VS. EXUDATE

Feature	Transudate	Exudate
Protein	Low	High
Specific gravity	< 1.012	> 1.020
Wall status	Intact	Leaky

QUICK TABLE · KILLING, OXYGEN-DEPENDENT VS. INDEPENDENT

Route	Key players
Oxygen-dependent	NADPH oxidase → O ₂ ⁻ → H ₂ O ₂ ; MPO + halide → HOCl
Oxygen-independent	Lysozyme, lactoferrin, defensins

QUICK TABLE · GIANT CELLS AT A GLANCE

Cell	Nuclei	Context
Langhans'	Peripheral horseshoe	Tuberculous granuloma
Foreign-body	Random/scattered	Inert foreign material

QUICK TABLE · DRUG TARGETS IN THE ARACHIDONIC-ACID PATHWAY

Drug class	Target	Effect
Corticosteroids	Phospholipase A ₂	Blocks both COX and LOX branches
Aspirin, NSAIDs	Cyclooxygenase	Blocks prostaglandins/thromboxane only; can shunt toward leukotrienes
Zileuton	5-lipoxygenase	Blocks leukotriene synthesis

Glossary

Alteration

Degeneration or necrosis of the local inflamed tissue or cell.

Chemotaxis

Directed leukocyte movement toward a chemical stimulus.

Chemotactic agent

A chemical that induces chemotaxis, e.g. C5a, LTB₄, bacterial peptides.

Complement

A plasma protein cascade activated by antigen-antibody complexes (classical) or microbial surfaces (alternative), converging on C3/C5 and the membrane attack complex.

Diapedesis

Synonym for transmigration: a leukocyte's passage through the vessel wall.

Edema

Excess fluid in interstitial tissue or serous cavities; may be transudate or exudate.

Exudate

Protein-rich, often cell-rich fluid that leaks through a permeable vessel wall in inflammation.

Exudation

Movement of intravascular fluid and cells into the interstitium, a cavity, or a surface.

Hageman factor (Factor XII)

Plasma factor that, once activated, triggers the kinin, clotting, and (indirectly) complement cascades together.

Hyperemia

Active, arteriolar-driven increase in blood within a tissue.

Margination

Leukocytes pushed out of the axial blood stream toward the vessel wall as flow slows.

Opsonin

A serum protein (chiefly IgG or C3b) that coats a particle to make phagocytosis easier.

Pavementing

Leukocytes lining the endothelial surface so densely it looks paved under the microscope.

Phagocytosis

Engulfment of a microorganism, foreign body, or debris by a leukocyte.

Phagolysosome

The vacuole formed when a phagosome fuses with a lysosome.

Proliferation (in inflammation)

Proliferation or regeneration of local tissue/cells: macrophages, fibroblasts, lymphocytes, endothelium, epithelium.

Rouleaux

Stacks of aggregated red cells formed as blood flow slows in stasis.

Stasis

Slowed, sludged blood flow following vasodilation, allowing margination and fluid loss.

Transudate

Low-protein, low-cell fluid that crosses an intact vessel wall due to hemodynamic imbalance.

Transmigration

A leukocyte's passage through the endothelial junction and basement membrane, driven by PECAM-1.